

P 6.3-1 + Determine the value of voltage measured by the voltmeter in Figure P 6.3-1.

Answer: -4 V

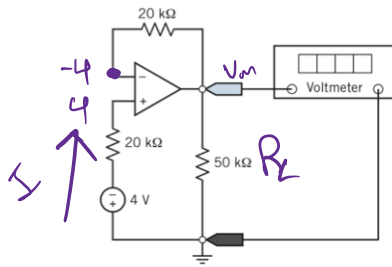


Figure P 6.3-1

The current out of negative side of amplifier is approx. zero.

Therefore, $20\text{ k}\Omega \times I \approx 0$

$$-4 - V_m = 0$$

$$V_m = -4\text{ V}$$

Home Work 5

P 6.3-3 + Find v_o and i_o for the circuit of Figure P 6.3-3.

Answer: $v_o = -30\text{ V}$ and $i_o = 3.5\text{ mA}$

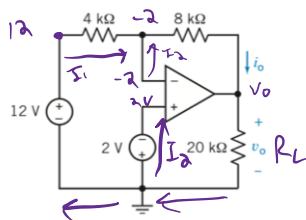


Figure P 6.3-3

KCL after $4\text{ k}\Omega$ resistor

$$I_1 = I_o - I_2$$

$$I_1 = 14/4\text{ k} = 3.5\text{ mA}$$

$$I_2 \approx 0\text{ mA}$$

$$I_o = 3.5\text{ mA}$$

$$-2 - v_o = 8 \times 3.5$$

$$-v_o = 28 + 2 = 30$$

$$v_o = -30\text{ V}$$

Problem #4

P 6.4-2 Find v_o and i_o for the circuit of Figure P 6.4-2.

Answer: $v_o = -4 \text{ V}$ and $i_o = 1.33 \text{ mA}$

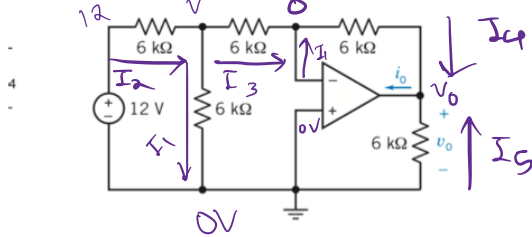


Figure P 6.4-2

Find

$$I_1 \approx 0$$

$$I_3 = I_4$$

$$I_4 = \frac{12 - v}{6k\Omega} - \frac{v}{6k\Omega} = I_3$$

$$I_3 = \frac{v}{6k} = \frac{12 - v}{6k\Omega} - \frac{v}{6k\Omega}$$

$$\frac{2v}{6k} = \frac{12 - v}{6k\Omega}$$

$$\frac{1}{3}v = 2 - \frac{1}{6}v \rightarrow v = 4 \text{ V}$$

$$I_4 = \frac{4}{6k} = \frac{2}{3} \text{ mA}, \text{ so}$$

$$0 - v_o = 6 \times \frac{2}{3} \quad v_o = -4 \text{ V}$$

@ v_o KCL $I_o = \frac{2}{3} \text{ mA} + \frac{0 - v_o}{5k} = \frac{4}{3} \text{ mA}$

Home Work 5

Problem #5

Non-Ideal

P 6.7-1 Consider the inverting amplifier shown in Figure P 6.7-1. The operational amplifier is a typical OP-07E (Table 6.7-1). Use the offsets model of the operational amplifier to calculate the output offset voltage. (Recall that the input v_{in} is set to zero when calculating the output offset voltage.)

Answer: 0.45 mV

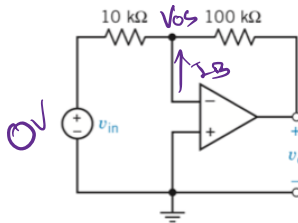


Figure P 6.7-1

• OP-07E offset - data

$$V_{os} = 30 \mu\text{V}$$

$$I_b = 1.2 \text{ nA}$$

KVL @ V_{os}

$$\frac{V_{os}}{10k} + \frac{V_{os} - v_o}{100k} + I_b = 0$$

$$\frac{v_o}{100k} = \frac{V_{os}}{10k} + \frac{V_{os}}{100k} + I_b$$

$$\begin{aligned} V_o &= \frac{30 \mu\text{V}}{10k} + \frac{30 \mu\text{V}}{100k} + 1.2 \text{ nA} \\ &= 3 \times 10^{-9} + 3 \times 10^{-3} + 1.2 \times 10^{-9} \\ &\quad \times 100k \\ &= 0.45 \text{ mV} \end{aligned}$$